



VISAYAS
STATE UNIVERSITY



Lesson 3.1 - Cartesian Coordinate System and Lines

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Objectives

At the end of the lesson, the students will be able to:

- Analyze and interpret the coordinate plane, including quadrants, axes, and points in two-dimensional space.
- Use the distance formula to find the distance between two points.
- Use the midpoint formula to find the midpoint of a line segment.
- Define and graph linear equations in two unknowns.
- Find the slope of a line.
- Define the forms of an equation of a line.
- Find an equation of a given line.

Cartesian Coordinate System

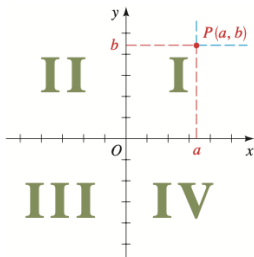


FIGURE 1

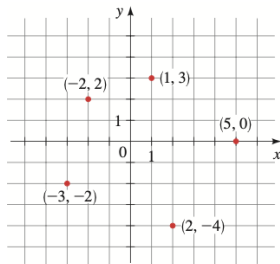
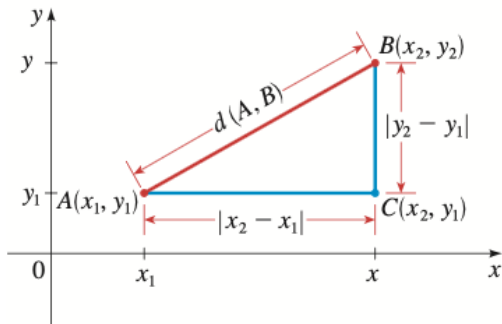


FIGURE 2

Terms to locate/identify/define: x -axis, y -axis, origin, quadrants, ordered pair.

The Distance Formula



Distance Formula: The distance between the points $A(x_1, y_1)$ and $B(x_2, y_2)$ in the plane is

$$d(A, B) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}.$$

The Distance Formula

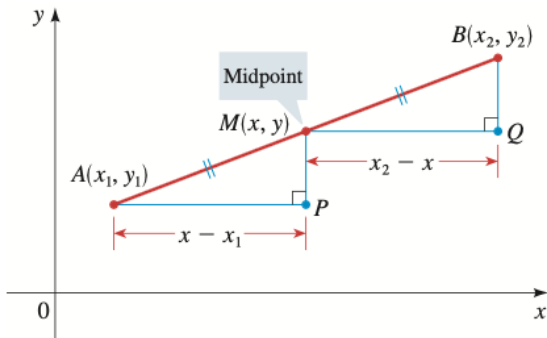
Example

Which of the points $P(1, -2)$ or $Q(8, 9)$ is closer to the point $A(5, 3)$?

Example

Show that the points $(2, 1)$, $(4, 0)$ and $(5, 7)$ are vertices of a right triangle.

The Midpoint Formula



Midpoint Formula: The midpoint of the line segment from $A(x_1, y_1)$ to $B(x_2, y_2)$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right).$$

The Midpoint Formula

Example

Find the midpoint of the line segment joining the points $(-5, -3)$ and $(9, 3)$.

Example

Show that the quadrilateral with vertices $P(1, 2)$, $Q(4, 4)$, $R(5, 9)$ and $S(2, 7)$ is a parallelogram by proving that its two diagonals bisect each other.

Linear Equations in Two Unknowns

Definition

An equation that can be reduced to the form $ax + by + c = 0$ where a, b, c are constants, and a and b , not zero is a **linear equation (two unknowns)** in x and y . If a or b is zero the equation is reduced to a **linear equation in one unknown**.

The **solution** of a linear equation in two unknowns is not unique since all the ordered pairs that satisfy the equation are solutions of the equation. Consider the equation

$$2x + y = 9,$$

the ordered pairs $(1, 7)$ and $(2, 5)$ are just two of the possible solutions of the equation. So we may denote the solution set as $\{(x, y) | 2x + y = 9\}$

Graph of Linear Equations in Two Unknowns

As its name suggests the graph of linear equation in two unknowns is a **straight line**. To graph a straight line we only need to know two distinct points called the x and y intercepts.

Definition

y -intercept:

- graphically, is the value on the y -axis where the line passes.
- algebraically, it is the value of y when $x = 0$.

x -intercept:

- graphically, it is the value on the x -axis, where the line passes.
- algebraically, the value of x when $y = 0$.

Graph of Linear Equations in Two Unknowns

Note 1: Only non-vertical lines have y -intercepts because vertical lines will not pass through the y -axis. **Note 2:** Only a non-horizontal line has an x -intercept.

Example

Determine the x and y intercept. Then draw the graph:

❶ $y = 4x + 5$

❷ $3x + 2y = 6$

❸ $3x = -15$

❹ $4y + 3x = 12 + 3x$

❺ $5y - 4x = 20$

❻ $4x = 2y + 16$

❼ $5y + 25 = -50$

❽ $3y = 9$

❾ $2x = 10$

❿ $y - 5/2x + 4 = 0$

Slope of a Line

The **slope** is one of the fundamentals of a line. It is the inclination of a line with respect to the x -axis. It is the ratio of the rise and run where the rise is the change in y and the run is the change in x .

Definition

Let m be the **slope** of a line, we have

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

where (x_1, y_1) and (x_2, y_2) are coordinates of two distinct points.

Example

For each pair of points, draw the line connecting the two points and find the slope of the line:

- 1 $(3, 8), (9, -4)$
- 2 $(-8, -7), (-9, -12)$
- 3 $(-16.3, 12.4), (-5.2, 8.7)$
- 4 $(17, -12), (-9, -15)$

Slope of a Line

The slope or steepness of a line can only be applied to nonvertical lines.

If the slope is **positive** the line **rise to the right** and when it's **negative** it implies that the line **rises to the left**. The **slopes of parallel lines are equal** while the **slopes of perpendicular lines are negative reciprocals** with each other or the product of their slopes is -1 . That is, $m_1 \cdot m_2 = -1$

Linear Equations in Two Unknowns

The **standard form** of linear equation in two variables is $ax + by + c = 0$ where a , b , and c are constants and a and b are not both equal to zero.

Definition (The Slope-Intercept Form)

The graph of the equation

$$y = mx + b$$

is a line whose slope is m and whose y -intercept is $(0, b)$.

Example

Sketch the graph of each linear equation.

① $y = 2x + 1$

② $y = 2$

③ $x + y = 2$

Forms of Linear Equations in Two Unknowns

Given an equation of a line, you can find its slope by writing the equation in slope-intercept form.

Example

Tell whether the graph of each pair of equations are parallel or perpendicular:

❶ $x + 6 = 4$, $y - x = -2$

❷ $2x + 5y = 8$, $6 + 2x = 5y$

❸ $5y = x - 7$, $5y = 3 + x$

❹ $3x + 2y = 6$, $3x - 2y + 1 = 0$

❺ $x = 6$, $6 - x = 8$

❻ $y + 3 = 5x$, $3x - y = -2$

Forms of Linear Equations in Two Unknowns

Definition (The Point-Slope Form)

The equation of the line with slope m passing through the point (x_1, y_1) is

$$y - y_1 = m(x - x_1).$$

- The point-slope form is most useful for finding the equation of a line. You should remember this form.
- The point-slope form can be used to find an equation of the line passing through two points (x_1, y_1) and (x_2, y_2) .

Definition (The Two-Point Form)

The equation of the line that passing through the points (x_1, y_1) and (x_2, y_2) is

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1} (x - x_1).$$

Forms of Linear Equations in Two Unknowns

Example

Find the equation of the line, given:

- ➊ $m = 4, (3, 2)$
- ➋ $m = -1/2, (8, 0)$
- ➌ $(1, 4) (5, 6)$
- ➍ $(0, 0) (7, 3)$
- ➎ $(3, 7)$ and parallel to $x + 2y = 6$
- ➏ $(-6, 2)$ and parallel to $3x - 9y = 6$
- ➐ $(2, 5)$ and perpendicular to $2x + y = -3$
- ➑ $(-3, -4)$ and perpendicular to $-3x + 6y = 2$

Applications

In real-life problems, the slope of a line can be interpreted as either a **ratio** or a **rate**. If the x -axis and y -axis have the same unit of measure, then the slope has no units and is a **ratio**. If the x -axis and y -axis have different units of measure, then the slope is a **rate** or **rate of change**.

Example

- 1 The maximum recommended slope of a wheelchair ramp is $\frac{1}{12}$. A business is installing a wheelchair ramp that rises 22 inches over a horizontal length of 24 feet. Is the ramp steeper than recommended?
- 2 A kitchen appliance manufacturing company determines that the total cost in dollars of producing x units of a blender is

$$C = 25x + 3500.$$

Describe the practical significance of the y -intercept and slope of this line.

Example (Straight-Line Depreciation)

A college purchased exercise equipment worth P600,000 for the new campus fitness center. The equipment has a useful life of 8 years. The salvage value at the end of 8 years is P100,000. Write a linear equation that describes the book value of the equipment each year.

Example (Predicting Sales per Share)

The sales per share for Starbucks Corporation were \$6.97 in 2001 and \$8.47 in 2002. Using only this information, write a linear equation that gives the sales per share in terms of the year. Then predict the sales per share for 2003.